



## Introduction

Research question : How are real minimum wages, real GDP per capita and unemployment rate associated with wage inequality across different segments of the wage distribution in Europe?

Wage inequality is a growing concern in European labour markets, shaped by differences in economic structures, institutions, and labour market conditions across countries. This study examines wage dispersion using interdecile ratios (D90/D10, D50/D10, D90/D50) to capture inequality at the bottom, middle, and top of the distribution. Using data for 21 EU countries with statutory minimum wages from 2018–2023, we analyse how real minimum wages, real GDP per capita, and unemployment relate to wage inequality across the distribution. Combining descriptive evidence with panel regression models, the study highlights asymmetric effects of macroeconomic conditions on different segments of the wage distribution.

## Data and Methodology

The analysis uses annual OECD data for 21 European countries with statutory minimum wages over the period 2018–2023. Wage inequality is measured using interdecile ratios (D90/D10, D50/D10, D90/D50). The main explanatory variables are the real minimum wage, real GDP per capita, and the unemployment rate. Panel regressions are estimated using fixed or random effects, selected via robust Hausman tests. Inference is based on cluster-robust standard errors at the country level to account for heteroskedasticity and within-country correlation.

## Results

The results show clear differences in how wage inequality behaves across the distribution. For the D90/D10 and D50/D10 ratios, robust Hausman tests favour fixed-effects models, while D90/D50 is better captured by a random-effects specification. For D90/D10, higher real GDP per capita and higher unemployment rates are associated with increased wage inequality, while the real minimum wage exhibits a small and statistically insignificant negative effect.

The D50/D10 ratio shows a similar pattern, suggesting that macroeconomic conditions play a stronger role in shaping inequality between the bottom and middle of the wage distribution than minimum wage policies.

For D90/D50, results are weaker and less robust: unemployment is associated with a reduction in inequality in the upper half of the wage distribution, while minimum wages and GDP per capita do not show statistically significant effects.

### Estimated Specifications (Selected Models)

**D90/D10 (Robust Fixed Effects)**  $D90/D10_{it} = -0.286 \log(\text{min\_wage}_{it}) + 1.207 \log(\text{gdp\_pc}_{it}) + 0.064 \text{unemployment\_rate}_{it}$

**D50/D10 (Robust Fixed Effects)**  $D50/D10_{it} = -0.202 \log(\text{min\_wage}_{it}) + 0.684 \log(\text{gdp\_pc}_{it}) + 0.043 \text{unemployment\_rate}_{it}$

**D90/D50 (Robust Random Effects)**  $D90/D50_{it} = -0.033 \log(\text{min\_wage}_{it}) - 0.153 \log(\text{gdp\_pc}_{it}) - 0.013 \text{unemployment\_rate}_{it}$

Coefficients reported from selected specifications based on robust Hausman tests. Inference is based on cluster-robust standard errors at the country level.

Table 1: Table 1: P-Values for Wage Dispersion (D90/D10)

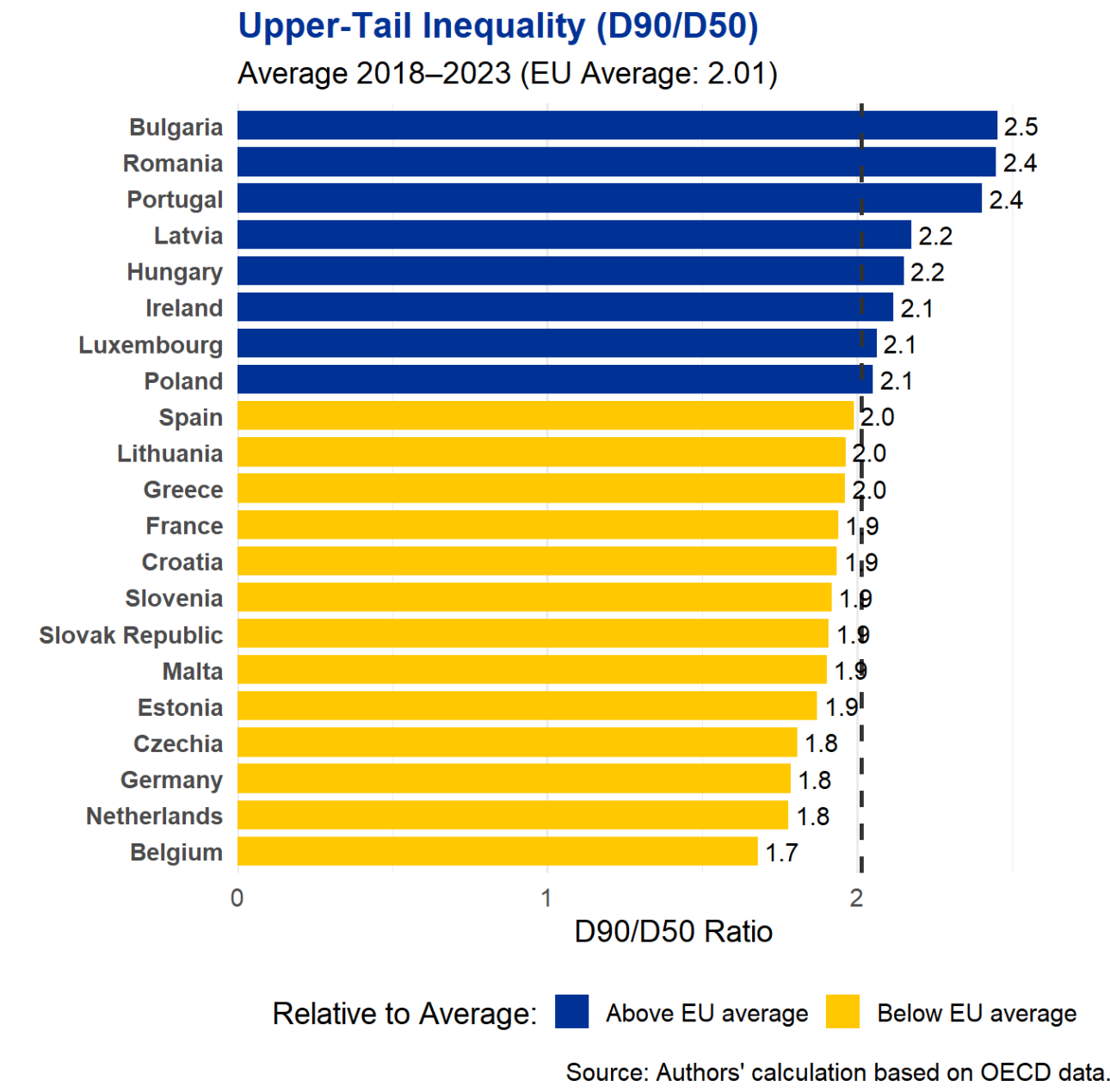
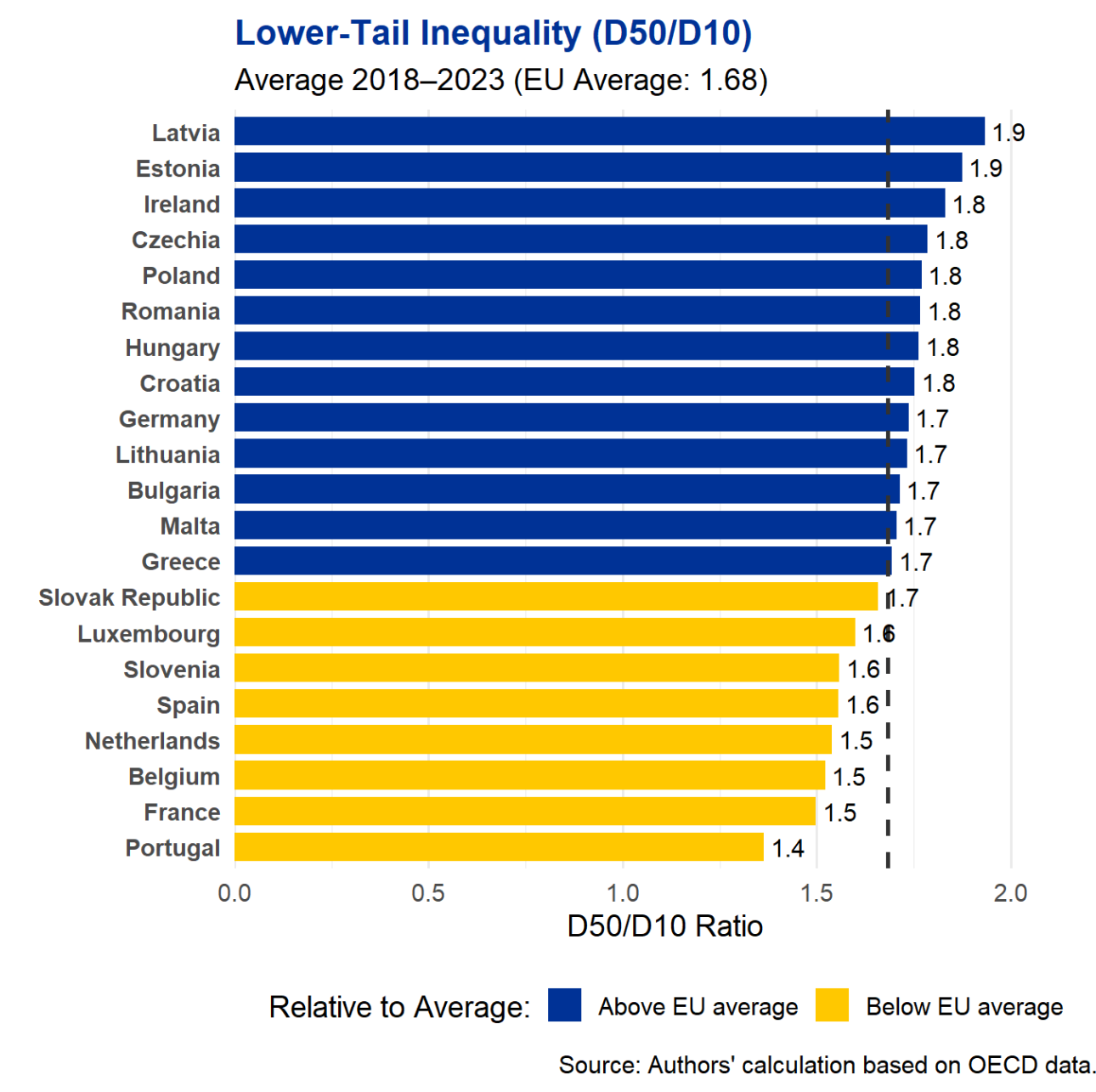
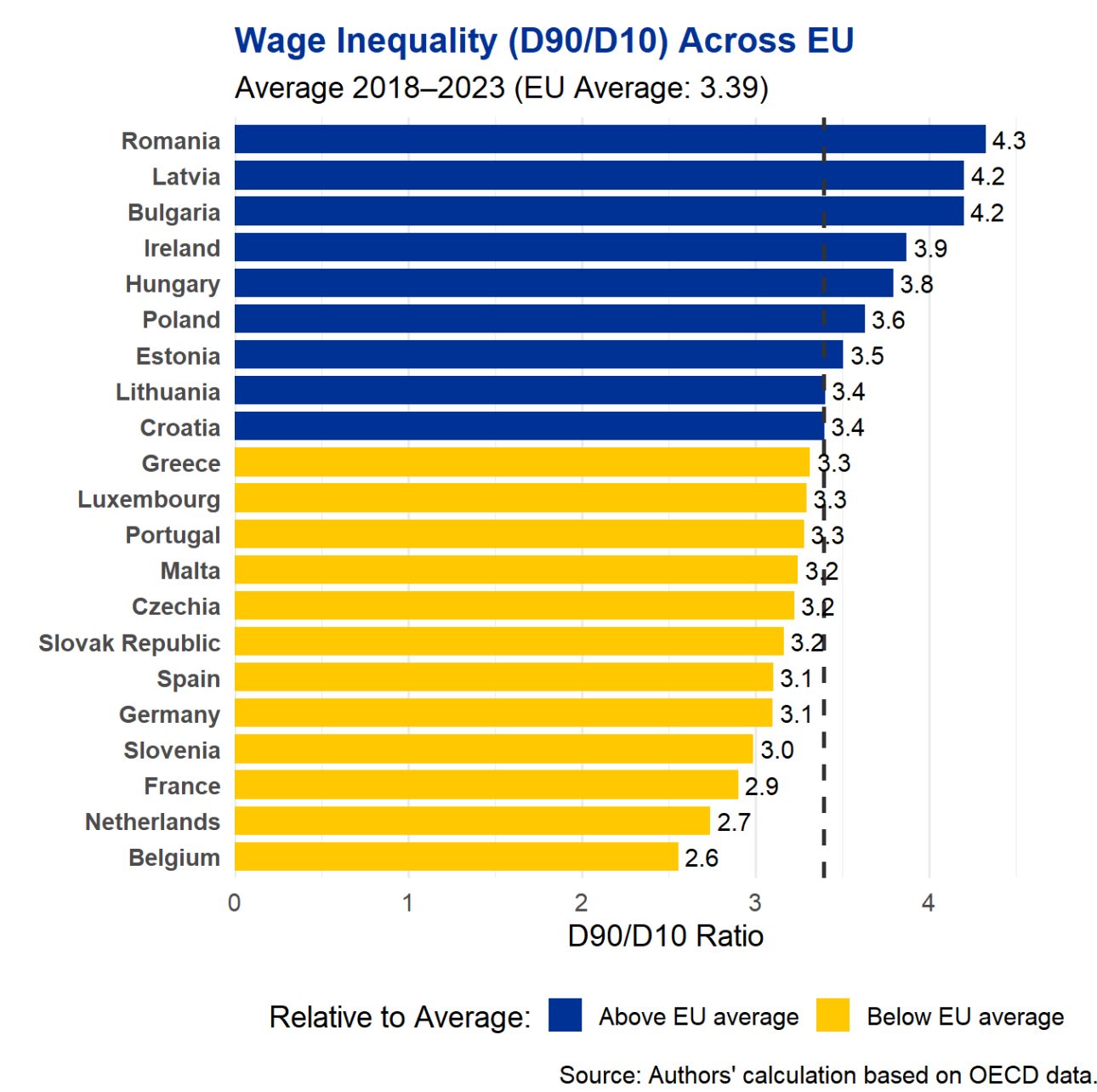
|                       | log_min_wage | log_gdp_pc  | unemployment_rate |
|-----------------------|--------------|-------------|-------------------|
| Pooled OLS            | 2.54e-10 *** | 7.53e-03 ** | 9.52e-01          |
| Fixed Effects         | 4.36e-01     | 8.12e-03 ** | 1.05e-02 *        |
| Random Effects        | 2.61e-03 **  | 1.20e-01    | 2.21e-01          |
| Robust Pooled         | 2.71e-04 *** | 2.16e-01    | 9.64e-01          |
| Robust Fixed Effects  | 3.77e-01     | 5.17e-03 ** | 3.50e-03 **       |
| Robust Random Effects | 1.63e-05 *** | 1.13e-01    | 9.38e-02 .        |

Table 1: Table 2: P-Values for Lower-Tail Inequality (D50/D10)

|                       | log_min_wage | log_gdp_pc   | unemployment_rate |
|-----------------------|--------------|--------------|-------------------|
| Pooled OLS            | 1.34e-09 *** | 1.11e-04 *** | 8.89e-01          |
| Fixed Effects         | 1.75e-01     | 2.58e-04 *** | 3.32e-05 ***      |
| Random Effects        | 2.70e-04 *** | 4.90e-03 **  | 4.23e-02 *        |
| Robust Pooled         | 5.05e-05 *** | 1.35e-02 *   | 9.23e-01          |
| Robust Fixed Effects  | 1.06e-01     | 1.15e-04 *** | 9.03e-07 ***      |
| Robust Random Effects | 1.39e-06 *** | 1.39e-03 **  | 4.07e-02 *        |

Table 1: Table 3: P-Values for Upper-Tail Inequality (D90/D50)

|                       | log_min_wage | log_gdp_pc | unemployment_rate |
|-----------------------|--------------|------------|-------------------|
| Pooled OLS            | 2.09e-03 **  | 8.84e-01   | 9.57e-01          |
| Fixed Effects         | 7.89e-01     | 3.00e-01   | 6.27e-02 .        |
| Random Effects        | 7.46e-01     | 1.61e-01   | 5.97e-02 .        |
| Robust Pooled         | 1.02e-01     | 9.49e-01   | 9.71e-01          |
| Robust Fixed Effects  | 7.14e-01     | 4.78e-01   | 1.89e-02 *        |
| Robust Random Effects | 6.68e-01     | 3.57e-01   | 2.91e-02 *        |



## Next Steps

Extend the analysis to countries without a statutory minimum wage. Incorporate additional institutional variables (e.g. collective bargaining coverage). Explore dynamic specifications to assess lagged effects over time.

## Conclusion

This study provides evidence that wage inequality in Europe varies markedly across different segments of the wage distribution. The results highlight a clear asymmetry: while real GDP per capita and unemployment rate are positively associated with higher inequality when the lower part of the distribution is involved (D90/D10 and D50/D10), these factors play a more limited role within the upper half of the distribution (D90/D50). In fixed-effects specifications, real minimum wages do not exhibit a statistically significant effect on wage inequality once country-specific heterogeneity is controlled for. This suggests that minimum wage policies are primarily associated with the lower tail of the wage distribution, without substantially altering broader patterns of wage dispersion. Overall, the findings indicate that macroeconomic conditions matter more for wage inequality than minimum wage levels, and that their impact differs across segments of the wage distribution. This underscores the importance of analysing inequality beyond single aggregate measures and adopting a distribution-sensitive perspective.

## References

OECD (2024). OECD Statistics Database. Organisation for Economic Co-operation and Development, Paris.

Autor, D., Manning, A., & Smith, C. (2016). The contribution of the minimum wage to U.S. wage inequality over three decades. American Economic Journal: Applied Economics, 8(1), 58–99.